

Mohammad Zhalechian

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Address: Kelley School of Business, ODT Department, 1309 E 10th St, Bloomington, IN 47405

APPOINTMENT

Indiana University, Bloomington, IN

2023 - Present

Assistant Professor of Operations and Decision Technologies

Kelley School of Business

Harvard University, Cambridge, MA

2022 - 2023

Postdoctoral Fellow

Harvard Kennedy School

EDUCATION

University of Michigan, Ann Arbor, MI

Ph.D. in Industrial and Operations Engineering (Operations Research)

2022

M.A. in Statistics

2019

University of Tehran, Tehran, Iran

M.Sc. in Industrial Engineering and Operations Research

2017

K. N. Toosi University of Technology (KNTU), Tehran, Iran

B.Sc. in Industrial Engineering and Operations Research

2014

RESEARCH INTERESTS

Big Data & Machine Learning, Human-AI Interfaces, Sequential Decision-Making;

Healthcare Delivery, Public Policy, Service Operations Management

PUBLICATIONS

Published Papers

[† denotes industry collaborators] [(M.D) denotes M.D. collaborators]

[1] **Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy**

Forthcoming at *Management Science*

M. Zhalechian, S. Saghafian, O. Robles[†]

- [Media Coverage: Transforming FDA Clearance: How AI and Human Insight Can Improve Medical Device Safety](#)
- [Media Coverage: A human-algorithm approach to approving medical devices could slash the recall rate and save billions](#)

[2] **Contextual Learning with Online Convex Optimization with Applications to Medical Decision-Making**

Management Science, 2025

Collaboration with Massachusetts Hospital

E. Keyvanshokoo, M. Zhalechian, C. Shi, MP. Van Oyen, P. Kazemian

- Finalist, CHOM Best Paper Competition (POMS), 2022
- Finalist, INFORMS MSOM Best Student Paper Competition, 2021
- Finalist, INFORMS Health Applications Society (HAS) Best Student Paper Competition, 2021
- Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper, 2020

[3] **Dynamic Care Unit Placements under Unknown Demand with Learning**
Manufacturing & Service Operations Management, 2025

Collaboration with St. Joseph Hospital
 A. Dean, M. Zhalechian, MP. Van Oyen

[4] **Data-driven Hospital Admission Control: A Contextual Learning Approach**
Operations Research, 2023

Collaboration with St. Joseph Hospital
 M. Zhalechian, E. Keyvanshokoo, C. Shi, MP. Van Oyen

- Winner of IOE Richard C. Wilson Best Student Paper Award on Service Systems, 2021
- Finalist, Seth Bonder Scholarship, INFORMS Health Applications Society, 2020

[5] **Online Resource Allocation with Personalized Learning**
Operations Research, 2022

Collaboration with Michigan Medicine
 M. Zhalechian, E. Keyvanshokoo, C. Shi, MP. Van Oyen

[6] **Validation of a Visual Field Prediction Tool for Glaucoma: A Multicenter Study Involving Patients with Glaucoma from Throughout the United Kingdom**
American Journal of Ophthalmology, 2025

Collaboration with Medisoft UK
 A. Dean, DJ. Fu (M.D), M. Zhalechian, MP. Van Oyen, M. Lavieri, AP. Khawaja (M.D), J. Stein (M.D)

[7] **Augmenting Kalman Filter Machine Learning Models with Data from Optical Coherence Tomography to Predict Future Visual Field Loss: The African Descent and Glaucoma Evaluation Study**
Ophthalmology Science, 2022

Collaboration with Kellogg Eye Center and Shiley Eye Institution
 M. Zhalechian, MP. Van Oyen, M. Lavieri, C. Andrew[†], J. Stein (M.D), et. al

Working Papers

[[†] denotes industry collaborators] [(M.D) denotes M.D. collaborators]

[8] **From Implementation to Impact: Human-Algorithm Collaboration in Hospital Bed Placement**

Under Review

Collaboration with Michigan Medicine

A. Dean, M. Zhalechian, I. Attari, M. Garifullin[†], J. Pardo[†], M. Van Oyen

- Implemented at Michigan Medicine Since July 2023
- Accepted at Michigan Medicine Quality Month Symposium, 2024

[9] **Does Early Pain Management Pay Off? Evidence on Utilization, Costs, and Surgeries**
 Under Review

Collaboration with Integrative Sports & Spine Pain Clinic at Southern California
 B. Li, M. Zhalechian, C. Chen, N. Vaughn (M.D), H. Lee (M.D)

[10] **Learning Bundle Pricing of Reusable Products**

Under Review

A. Dean, M. Zhalechian, C. Shi

- [Finalist, Service Science Best Student Paper Competition, 2024](#)

Selected Published Papers Prior Ph.D.

– **Sustainable Design of a Closed-loop Location-routing-inventory Supply Chain Network Under Mixed Uncertainty**

Transportation Research Part E, 2016

M. Zhalechian, R. Tavakkoli-Moghaddam, B. Zahiri, M. Mohammadi

– **Hub-and-spoke network design under operational and disruption risks**

Transportation Research Part E, 2017

M. Zhalechian, SA. Torabi, M. Mohammadi

– **A Self-adaptive Evolutionary Algorithm for a Fuzzy Multi-objective Hub Location Problem: An Integration of Responsiveness and Social Responsibility**

Engineering Applications of Artificial Intelligence, 2017

M. Zhalechian, R. Tavakkoli-Moghaddam, Y. Rahimi

HONORS AND AWARDS

(* denotes honors or awards received by students)

2026 Kelley School of Business Research Award

2024* Finalist, Service Science Best Student Paper Competition

2022 Winner, Towner Prize for Distinguished Academic Achievement

Rackham Graduate School, University of Michigan

2022 Winner, Rackham Outstanding Graduate Student Instructor Award

Rackham Graduate School, University of Michigan

2022 Finalist, CHOM Best Paper Competition (POMS)

2021 Rackham Graduate Research Grant

Rackham Graduate School, University of Michigan

2021 Winner, Richard C. Wilson Best Student Paper Award

IOE Department, University of Michigan

2020 Second Place, INFORMS Decision Analysis Society (DAS) Best Student Paper

2020 Finalist, INFORMS Seth Bonder Scholarship for Applied OR in Health Services

2020 Winner, Joel and L. Brown Graduate Student Instructor of the Semester

IOE Department, University of Michigan

2018-2021 Rackham Travel Grant Awards

2017 Graduate Student of the Year

College of Engineering, University of Tehran

Harmonizing Safety and Speed: A Human-Algorithm Approach to Enhance the FDA's Medical Device Clearance Policy

- Wayne State University, ISE Seminar Series *Nov. 2025*
- INFORMS Annual Meeting, Georgia, Atlanta *Oct. 2025*
- INFORMS MSOM, London, UK *Jun. 2025*
- INFORMS Annual Meeting, Seattle, Washington *Oct. 2024*
- POMS Annual Conference, Minneapolis, Minnesota *Apr. 2024*
- INFORMS Annual Meeting, Phoenix, Arizona *Oct. 2023*
- INFORMS Healthcare, Toronto, Canada *Jul. 2023*

Interpretable Design of Multi-Agent Mobile Health Applications

- INFORMS Healthcare, Toronto, Canada *Jul. 2023*

Data-Driven Learning and Resource Allocation in Service Operations Management

- Dartmouth College, Tuck School of Business *Jan. 2023*
- Indiana University, Kelley School of Business *Jan. 2023*
- Carnegie Mellon University, Tepper School of Business *Dec. 2022*
- Rice University, Jones Graduate School of Business *Dec. 2022*
- University of North Carolina at Chapel Hill, Kenan-Flagler Business School *Dec. 2022*
- University of Illinois Urbana-Champaign, Gies College of Business *Dec. 2022*
- Miami University, Herbert Business School *Dec. 2022*
- University of Illinois Chicago, College of Business Administration *Dec. 2022*
- Baruch College, Zicklin School of Business *Dec. 2022*

Personalized Hospital Admission Control: A Contextual Learning Approach

- INFORMS Annual Meeting, Indianapolis, IN *Oct. 2022*
- POMS Annual Conference, Online *Apr. 2022*
- INFORMS Annual Meeting, Online *Oct. 2021*
- POMS Annual Conference, Online *May 2021*
- INFORMS Annual Meeting, Online *Nov. 2020*
- INFORMS Annual Meeting, Seattle, WA *Oct. 2019*
- INFORMS Healthcare Conference, Cambridge, MA *July 2019*
- POMS Annual Conference, Washington, DC *May 2019*

Online Advance Appointment Scheduling with Personalized Learning

- INFORMS Annual Meeting, Online *Oct. 2021*
- INFORMS Annual Meeting, Online *Nov. 2020*

Joint Learning and Optimization for Healthcare Operations and Medical Decision-Making

- INFORMS Annual Meeting, Online *Nov. 2020*
(Seth Bonder Scholarship Session)

Kalman Filtering Based Machine Learning to Predict Future Mean Deviation Values for Patients with Glaucoma

- ARVO Annual Meeting 2020, Online *May 2020*

A Distributionally Robust Capacity Planning Model for Optimizing Access Delay in Surgical Services under Limited Information

- INFORMS Healthcare, Cambridge, MA *July 2019*
- INFORMS Annual Meeting, Phoenix, AZ *Nov. 2018*

TEACHING EXPERIENCE

Instructor, Indiana University, Bloomington, IN *Spring 2024, 2025*
Class: *Business Analytics and Modeling* (K353)

Instructor, University of Michigan, Ann Arbor, MI *Fall 2020*
Class: *Economic Decision Making* (IOE 201)

Recitation Instructor, University of Michigan, Ann Arbor, MI *Winter 2020*
Class: *Introduction to Markov Processes* (IOE 316)

Teaching Assistant, University of Michigan, Ann Arbor, MI

- Practicum in Hospital Systems (IOE 481) *Fall 2021-Winter 2022*
- Service Operations Management (IOE 419) *Winter 2021*
- Probability and Statistics for Eng. (IOE 265) *Fall 2018-Winter 2019- Fall 2019*

ACADEMIC SERVICE

Award Committees

- Judge, Public Sector OR Best Paper Competition, 2023

PhD Thesis Committees

- Ali Kaan Kurbanzade, Kelley School of Business, 2025-present
- Iman Attari, Kelley School of Business, 2023-2025. Placement: Carl H. Lindner College of Business, University of Cincinnati

Recruiting Committees

- PhD Recruiting Committee, Kelley School of Business, Operations and Decision Tech., 2025
- Faculty Recruiting Committee, Kelley School of Business, Operations and Decision Tech., 2024
- Faculty Recruiting Committee, University of Michigan, Industrial and Operations Engineering (Graduate Student Committee), 2021

Referee for Journals

- Management Science
- Operations Research
- Manufacturing & Service Operations Management

- Production and Operations Management
- Information Systems Research
- European Journal of Operational Research
- International Journal of Management Science (Omega)
- Health Care Management Science

Session Organizer & Chair for Conferences

- INFORMS Annual Meeting, 2026
(AI in Healthcare)
- INFORMS Healthcare Conference, 2026
(AI for Healthcare Oversight and Operations & AI and Analytics for Decision-Making in Healthcare)
- INFORMS Annual Meeting, 2025
(Data-driven Healthcare Analytics)
- POMS, 2024
(Policies and Regulation in Healthcare)
- INFORMS Annual Meeting, 2021
(Sequential Decision Making and Machine Learning in Healthcare)
- POMS, 2021
(Healthcare Operations Management)
- INFORMS Annual Meeting, 2020
(Predictive Analytics & Sequential Decision-Making in Healthcare)